What Makes Neural Networks Easy to Fool? A Controlled Tour of the Candidate Root Causes

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DRAFT

Abstract

Networks are now far more robust to small, deliberate changes than they were a few years ago, yet there is still no agreed answer to the basic question: *what is the core issue?* Many explanations have been proposed: networks add up many tiny effects, they rely on predictive but fragile features, they prefer texture to shape, their models are not sparse, they ignore directions the data never explores, they have too little data, or training simply finds the wrong solution. They are rarely compared on equal terms. This paper is an illustrative study for junior researchers and curious readers. We test each explanation on its own, in a small problem where the truth is known, with one yardstick throughout: the smallest change that flips the model's answer, and how it compares with the best that any model could do. We organise the findings around one split. Some vulnerability is *inevitable*: given the data and the allowed change, no model can be both accurate and robust. The rest is *avoidable*: a better model exists, but the objective, the optimiser, the architecture or the data do not find it. In our toy worlds, the dominant effect is the inevitable one, namely evidence spread over many inputs under a per-input budget. It also accounts for shape versus texture: for the best possible rule only how widely a cue is spread matters, not its frequency. On top of it we find avoidable effects that are real but smaller: ignoring irrelevant inputs (up to $2\times$), a convolutional network that reads fine textures badly, optimiser bias towards the wrong kind of solution, and sensitivity off the data manifold, which disappears with weight decay. A network trained only on adversarial changes still learns the true classes, so the changes are made of real features. More data, in these experiments, does not help under standard training. Real-data confirmation is still being run and is marked in this draft. It is an aid to intuition and a map of causes, not a claim about large networks in general.

1 The puzzle and a way to organise the answer

A faint pattern of noise, invisible to a person, makes an accurate image classifier change its answer. Adversarial training and other defences now make models much more robust, but we still lack a clear account of *why* ordinary models are fragile in the first place. The literature offers many candidate causes; each is usually shown on its own, on a different dataset, with a different measure.

We do two things. First, we compare candidate causes in the same small, fully controlled worlds with the same measure. Second, we sort what we find using one distinction.

- **Inevitable (data-limited).** Given the data and the allowed change ϵ, no classifier can be both accurate and robust. The classes overlap within ϵ, or the evidence is spread so thinly that robustness costs accuracy.

- **Avoidable (learning-limited).** A classifier that is accurate *and* robust exists, but the training objective, the optimiser, the architecture or the amount of data does not find it.

Every hypothesis below is a claim about which part it explains. A cause that is *necessary* must appear whenever vulnerability does; a cause that is *sufficient* produces vulnerability by itself; a cause may also only *contribute*. Our scorecard (Section 5) records which, with a plain statement of how strong the evidence is.

The yardstick. For every model and every correctly classified test point we compute the *flip size*: the smallest change that flips the answer. We measure it two ways: the largest change to any single input (per-input, ℓ_∞) and the length of the whole change (total, ℓ_2). Their ratio tells how widely the model spreads its attention. Where the data are generated by us, we also compute the same quantity exactly for the best possible rule.

How to read the paper. Each subsection asks one question, runs one small experiment, and ends with a boxed verdict. Boxes labelled "TO DO" mark experiments still running. Settings are in Appendix A; unless stated, numbers are means over 10 runs.

2 The toy world

Every example is six numbers and a label 0 or 1 with equal chance. The first number is a *reliable clue* (its class averages are 1.0 apart; looking at it alone gives 95% accuracy). The other five are *weak clues* (class averages 0.4 apart; each alone gives 79%). The noise is bell-shaped and the values are not clipped, so the best possible rule is a weighted sum of the six numbers, answered by its sign. It reaches 99.3%. Five weak clues together reach 96.3%, better than the reliable clue alone. A small network (one hidden layer of 8 units) trained on this task reaches the same accuracy.

For a weighted sum with weights w_i and total $\sum_i w_i x_i$, an attacker who may change every input by at most ϵ moves the total by $\epsilon \sum_i |w_i| = \epsilon \|w\|_1$. The answer flips when this exceeds the answer's confidence, so

$$\text{per-input flip size} = \frac{\text{confidence}}{\|w\|_1}, \qquad \text{total flip size} = \frac{\text{confidence}}{\|w\|_2}.$$

This one formula organises most of what follows. It is exact for the best rule; for trained networks we find the flip size by search.

3 The tour

3.1 H1: evidence added up over many inputs

Claim (Goodfellow et al.): a small change to every input, all in the unhelpful direction, adds up. **Experiment:** keep the total evidence fixed and split it equally over $k = 1, 2, \ldots, 64$ inputs, so accuracy stays at 95%.

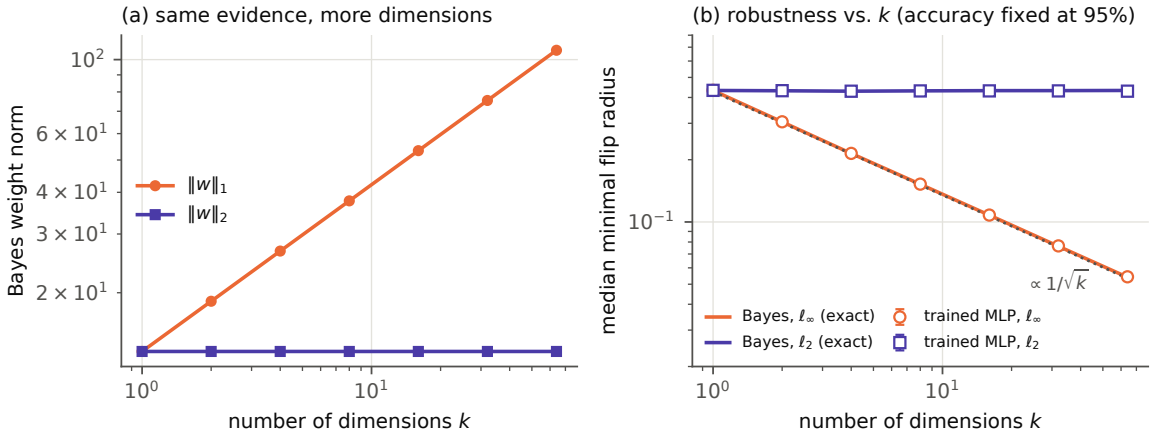


Figure 1: Same evidence spread over k inputs. Per-input flip size (steep lines) falls as $1/\sqrt{k}$; total flip size (flat lines) does not move. Exact best rule (lines) and trained networks (markers) coincide.

The per-input flip size falls from 0.43 at $k = 1$ to 0.054 at $k = 64$ (a factor of eight), with the same accuracy and the same total flip size (0.43). The trained network lies on the exact curve.

Verdict. Necessary and sufficient, in this world, for fragility to small per-input changes. It is *inevitable*: the best rule has it too. It disappears if the attacker is limited by total size (ℓ_2) instead, so it is a statement about the pairing of the data with the allowed change.

3.2 H2: small clues, not unreliable ones

Claim (Ilyas et al.): networks rely on features that are predictive but fragile. **Experiment:** one clue; vary its size (gap between class averages) and its reliability independently.

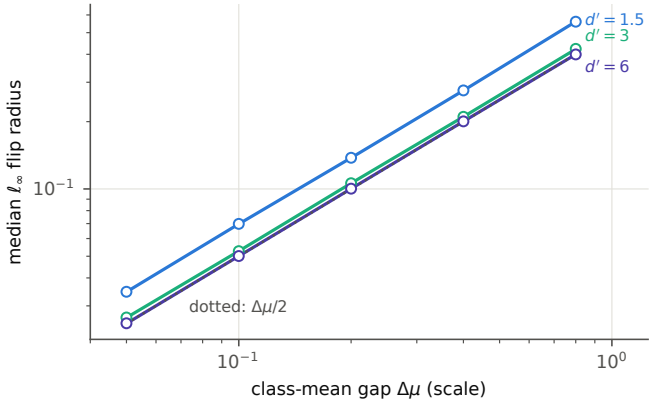


Figure 2: Flip size of a single clue against its size, at three reliabilities. The lines nearly coincide: size matters, reliability hardly does.

Flip size is proportional to the clue's size. Raising reliability from 77% to 99.9% changes it by only about 1.4×, and in the wrong direction for the usual intuition: a more reliable clue is slightly *easier* to flip. A clue that is 99.9% reliable but tiny (gap 0.05) flips at 0.025.

Verdict. "Non-robust" means small relative to the allowed change, not unreliable. Inevitable (a property of the data). Still to do: the transfer test (do changes crafted against a small-scale-feature model fool a large-scale-feature model?).

Transfer test: are adversarial changes made of real features? Ilyas et al. showed that a network trained only on adversarially changed images, with labels given by the *target* of the change, can still classify clean images. We repeat this in miniature. For each training point we craft a targeted change of size ϵ towards a randomly chosen target label and keep the changed input with that target label. The labels are random with respect to the true class, so a fresh network can do well on clean data only if the changes themselves carry features.

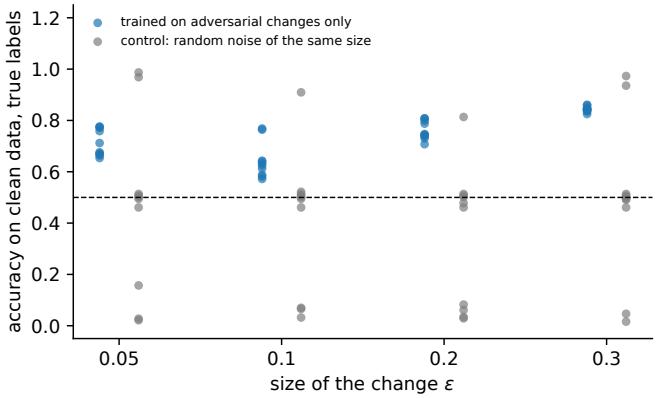


Figure 3: Accuracy on clean data (true labels) of a fresh network trained only on adversarially changed inputs with random target labels (blue, 10 runs), against a control trained on random-sign noise of the same size (gray).

The fresh network is above chance in every one of the 10 runs at every ϵ (mean 0.71, 0.64, 0.76, 0.84 for $\epsilon = 0.05, 0.1, 0.2, 0.3$). The control scatters between 0.02 and 0.99: an almost untrained network lines up with the reliable clue by chance, positively or negatively, and shows no consistent signal.

Verdict. In this world an adversarial change is built from the data's own predictive directions. This supports the "features, not bugs" reading in its simplest form. It is unsurprising for a weighted sum, which is the point: very little needs to be assumed.

3.3 H4: sparsity

Sparsity can mean three different things. We test each.

(a) Sparse versus dense signal. Put the same total evidence in s of 64 inputs; the other $64 - s$ inputs are pure noise.

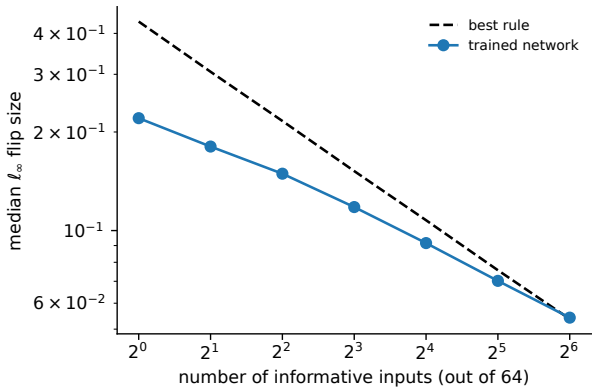


Figure 4: Median per-input flip size against the number of informative inputs (out of 64), same total evidence. Dashed: best rule. Blue: trained network.

The best rule's flip size falls as $1/\sqrt{s}$ (from 0.43 at $s = 1$ to 0.054 at $s = 64$): a dense signal is easier to fool. But the trained network is about *twice as fragile* as the best rule when the signal is sparse (0.22 against 0.43 at $s = 1$) and matches it when the signal is dense. The network does not fully ignore the irrelevant inputs.

(b) Sparse weights. Penalties that switch whole inputs off (first-layer L_1, group Lasso, elastic net) cut the attack on the weak clues from 0.260 to 0.02–0.05 at an accuracy cost of 1.5–3 points. Ordinary weight shrinkage (L_2) does not: it helps a little at no cost, then collapses to chance at strength 0.4.

(c) Sparse activity. Penalising hidden activity lowers activity (0.42 $\rightarrow$ 0.07) but leaves the weight pattern (1.97 $\rightarrow$ 1.81) and the attack (0.26) essentially unchanged, and it fails all at once: each run either stays accurate or becomes a dead network (0 of 10 runs at strength 2, all 10 at strength 8).

Verdict. Sparsity of the *signal* is not a cause of fragility; *density* is (it is H1 again). What the model does with irrelevant inputs is an *avoidable*, learning-limited contribution of up to 2$\times$ here. Sparsity of the *weights* removes fragility at a price (accuracy); sparsity of the *activity* does not help.

3.4 H8: random noise versus the worst case

Claim (Ford et al.): adversarial examples are the far tail of ordinary sensitivity to noise. **Experiment:** for the best rule, compare the size of random $\pm$ noise needed to flip a quarter of the points with the size of the worst-case change.

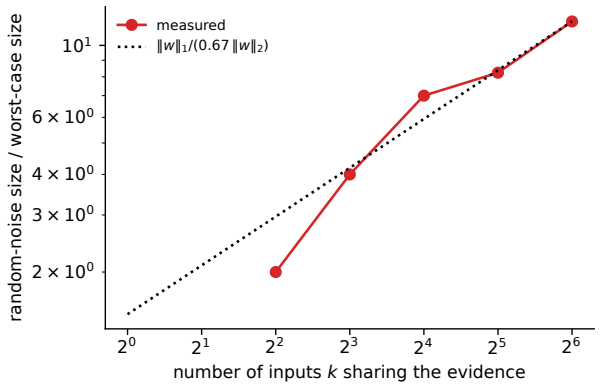


Figure 5: How many times larger random noise must be than the worst case to flip the answer, against the number of inputs k sharing the evidence. Dotted: $\|w\|_1/(0.67\|w\|_2)$.

The gap grows like $\sqrt{k}$ (about 2 at $k = 4$, 12 at $k = 64$) and follows $\|w\|_1/\|w\|_2$.

Verdict. Not a separate cause but the same one seen from another side: the more inputs share the evidence, the more the worst case beats random noise. No training is involved.

3.5 H10: the inevitable ceiling and the avoidable gap

Here we ask directly how much vulnerability is inevitable. For a linear rule with weights w the accuracy under the worst change of size ϵ is exactly $\Phi\big((\frac{1}{2}\sum_i w_i\Delta\mu_i - \epsilon\|w\|_1)/\sqrt{\sum_i w_i^2\sigma_i^2}\big)$ ($\Delta\mu_i$ the class gap, σ_i the noise of input i). We maximise it over w at each ϵ to get the best accuracy under attack that any linear rule can reach.

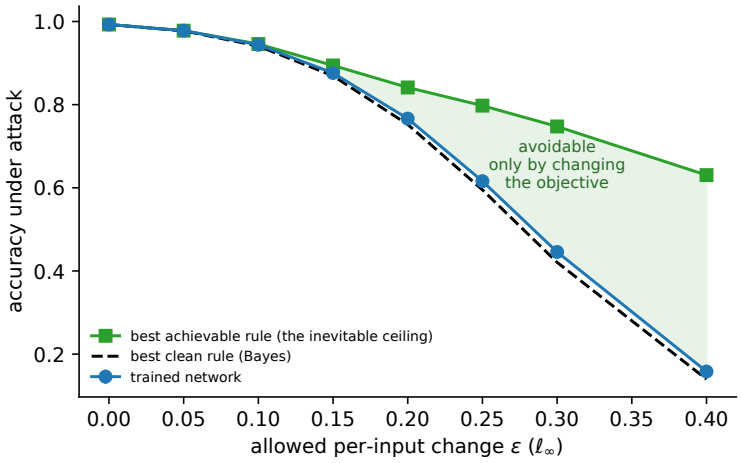


Figure 6: Accuracy under attack against the allowed per-input change. Green: the best achievable rule (the inevitable ceiling). Blue: a trained network. Dashed: the best clean rule. The shaded gap is avoidable, but only by changing the training objective.

At $\epsilon = 0.3$ the ceiling is 0.75 (reached by using the reliable clue only), while the best clean rule gets 0.42 and the trained network 0.45. The ceiling itself falls as ϵ grows (0.63 at $\epsilon = 0.4$): part of

the loss is inevitable. The gap between ceiling and network, about 0.3 at $\epsilon = 0.3$, is avoidable, yet *standard training will not close it*, because the standard objective rewards the clean-optimal rule. For small ϵ (≤ 0.1) the two coincide.

Verdict. Both parts are real. The ceiling is inevitable. The gap is avoidable by changing the objective, which is the accuracy/robustness trade-off in numbers. We do not study adversarial training here. The ceiling is for linear rules only.

3.6 H7: which solution the optimiser reaches

Gradient descent on separable data is known to head for the solution with the largest margin measured in ℓ_2, not in ℓ_∞. **Experiment:** 100 inputs, s of them informative, 60 training examples (so the data are separable). Compare the ℓ_2 max-margin classifier (we use an SVM as the stand-in for the gradient-descent limit) with the ℓ_∞ max-margin classifier (a linear programme) and the oracle best rule.

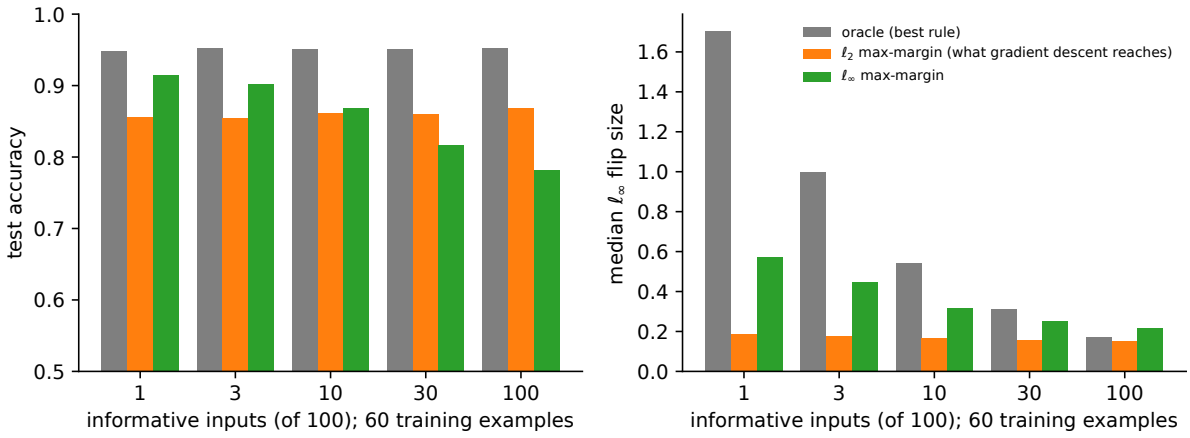


Figure 7: Accuracy (left) and per-input flip size (right) of three linear classifiers, against the number of informative inputs. The ℓ_∞ max-margin solution beats the ℓ_2 one when the signal is sparse.

With a sparse signal ($s = 1$) the ℓ_∞ solution is better on both axes (accuracy 0.92 against 0.86; flip size 0.57 against 0.19). With a dense signal the gap closes, and the ℓ_∞ solution then trades accuracy (0.78 against 0.87) for a larger flip size (0.22 against 0.15).

We checked the stand-in directly: 20 000 steps of gradient descent on the logistic loss from zero end in a direction with cosine similarity 0.98 to the SVM solution, with accuracy and flip size within about 0.015 of it (for $s = 1$: accuracy 0.869 against 0.855, flip size 0.199 against 0.186).

Verdict. The optimiser's bias towards ℓ_2 is an avoidable contribution when the truth is sparse and data are scarce. When the truth is dense, the gap is the trade-off again.

3.7 H6: more data

Claim (Schmidt et al.): robust generalisation needs more data. **Experiment (standard training only):** train on $n = 30$ to 4000 examples, with the evidence spread over 16 inputs.

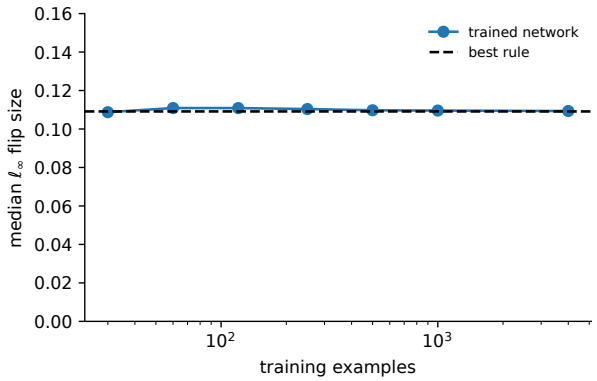


Figure 8: Per-input flip size against the number of training examples. The trained network matches the best rule at every n.

Accuracy rises from 0.905 to 0.953, but the flip size equals the best rule's (0.109–0.111) at every n.

Verdict. In this world, more data under standard training does not change vulnerability. Caveat: the network is strongly regularised (weight decay), which may hide an effect; and we have not trained adversarially, where data quantity is known to matter. Not supported as a cause here.

3.8 H5: directions the data never use

Claim (Gilmer et al.; Stutz et al.): networks are sensitive in directions off the data manifold, where training gives no signal. **Experiment:** data lie on a 4-dimensional subspace of a 64-dimensional space (plus small off-subspace noise). Attack only along the subspace, or only across it.

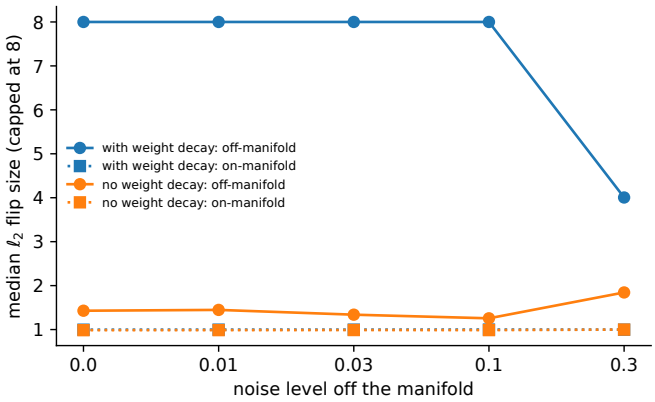


Figure 9: Median total flip size when the attack is limited to the data subspace (squares) or to the directions across it (circles), with and without weight decay. Values above 8 are shown as 8 (no flip found).

With weight decay the network ignores off-manifold directions entirely (no flip found within 8). Without weight decay, off-manifold flip sizes are 1.3–1.8, close to the on-manifold 1.0, and an attack free to use both is about 20% stronger than one limited to the subspace.

Verdict. Real but modest, and *avoidable*: the best rule is blind to these directions and ordinary weight decay makes the network blind too. Contributes; not the main cause here.

3.9 H3: shape versus texture

Claim (Geirhos et al.): networks lean on texture (fine, high-frequency patterns) more than people do, and texture reliance makes them fragile. **World:** 16×16 images with Gaussian pixel noise. A cue is a fixed pattern added with a class-dependent sign; its strength, and hence the accuracy it supports ($\approx 95\%$), is set exactly.

Experiment 1: frequency or spread? Four cues of equal strength: *smooth* or *fine* (a checkerboard), each *spread* over the whole image or *concentrated* in the middle. We report the cue's spread as the ℓ_1/ℓ_2 ratio of its pattern (14.4, 7.1, 16.0, 7.1).

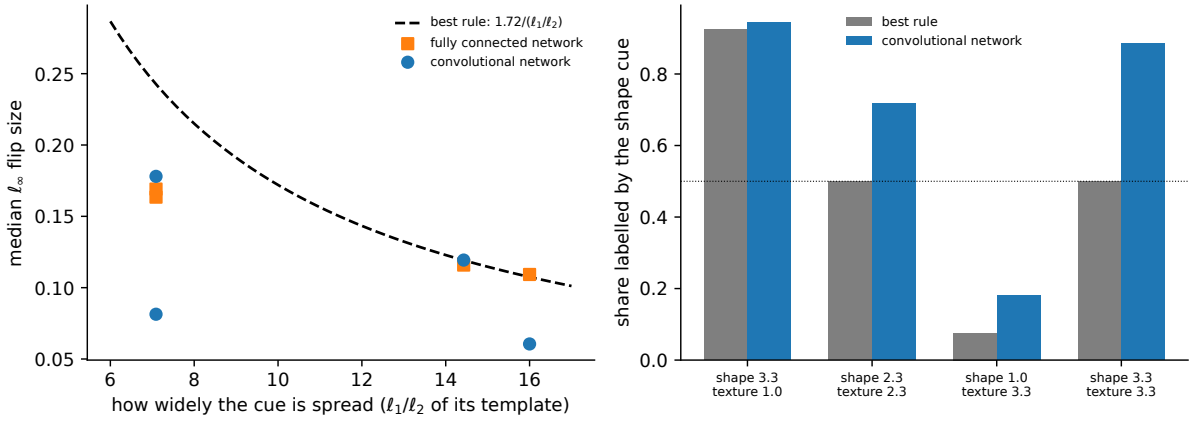


Figure 10: Left: per-input flip size of the best rule (dashed), a fully connected network (squares) and a convolutional network (circles) for four cues of equal strength, against how widely the cue is spread. Right (experiment 2): in images where the shape cue and the texture cue disagree, the share labelled by the shape cue, for the best rule and the convolutional network.

For the best rule, flip size times spread is the same for all four cues (1.70–1.73, equal to the total flip size): *frequency does not matter; spread does.* The fully connected network follows the best rule on the spread cues (0.116 against 0.120; 0.109 against 0.107) and is 1.45× more fragile on the concentrated ones (0.163 against 0.240; 0.169 against 0.242), the same effect as in H4a. The convolutional network (two convolution–ReLU–max-pool stages) matches the best rule on the smooth cues but not on the fine ones: accuracy falls to 0.87 and 0.86 and the flip size to 0.06 and 0.08 (best rule 0.11 and 0.24), and even the total flip size drops (0.72 against 1.7).

Experiment 2: cue conflict. Images carry both a shape-like cue (smooth, concentrated) and a texture-like cue (fine, spread). In test images the two disagree. The best rule follows whichever cue is stronger, and is indifferent at equal strength. The convolutional network follows the stronger cue too, but at equal strength it is *shape-biased*: it follows the shape cue in 72% and 89% of images for equal strengths 2.3 and 3.3. Flip sizes follow what the model relies on: 0.19 when the shape cue dominates (strengths 3.3 and 1.0) and 0.067 when the texture cue does (1.0 and 3.3), 2.8× more fragile (the spread ratio is 2.3). A worst-case change puts its energy along the cue the model

uses (fraction along the shape template 0.39 against 0.03 along the texture template when shape dominates; 0.10 against 0.33 when texture dominates).

Verdict. "Texture" as a category is not a separate cause. For the best rule only spread matters, so a fine pattern is fragile when it is spread over many pixels and not otherwise. Trained networks add architecture-specific effects: this convolutional network reads fine patterns badly (avoidable) and prefers shape at equal strength. Caveat: a 16×16 toy and one architecture; the real-data check is still to do.

3.10 H9: architecture and invariance

Question: does building in the right invariance reduce vulnerability, or only move it? **World:** $16\times$ 16 noisy images; the label is whether a small known pattern (3×3 block) is present *anywhere*. We compare a fully connected network, a small translation-invariant convolutional network (convolution plus a global maximum), and the hand-built matched filter, which knows the pattern.

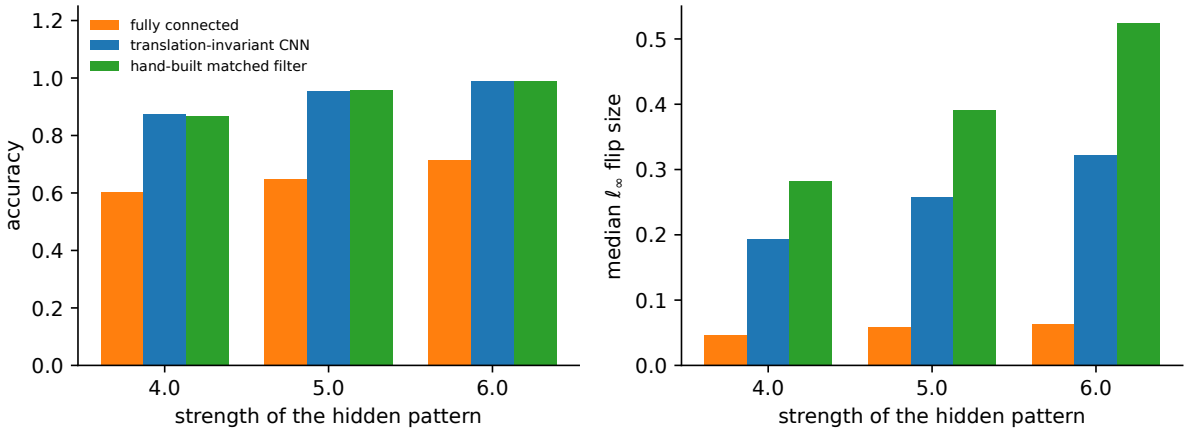


Figure 11: Accuracy (left) and per-input flip size (right) against the strength of the hidden pattern, for three models (3 runs).

The right invariance helps on both axes: at strength 6 the convolutional network reaches accuracy 0.99 with flip size 0.32, while the fully connected network (data-starved with 3000 examples) reaches 0.72 with 0.064. But the learned convolutional network is still about $1.6\times$ more fragile than the matched filter at the same accuracy (0.32 against 0.52 at strength 6; 0.19 against 0.28 at strength 4). We have not diagnosed why.

Verdict. The right inductive bias reduces vulnerability substantially, but a learned solution does not reach the best hand-built one at equal accuracy: an avoidable gap remains. Caveat: the fully connected network is not at equal accuracy, so its flip size mixes data hunger with fragility.

4 Do the toy findings survive on real images?

TO DO (experiment not finished). Systematic checks per hypothesis on MNIST, Fashion-MNIST and CIFAR-10.

Two checks already exist. Mixing the pixels of MNIST digits with a random rotation (same information, same total distances) leaves total flip size unchanged but shrinks per-input flip size by only 12–14%, not the eightfold drop of the toy world: pixel data are already in the spread-out regime. On CIFAR-10, letting the network use more principal components raises accuracy (44% → 78%) and shrinks *both* flip sizes by about 7–8×, so the loss there is a shrinking margin, not spreading. A pilot comparing the networks with nearest-neighbour classifiers shows why "a robust classifier exists" is hard to establish in ℓ_∞: on raw pixels every pair of images is almost equally far apart (nearest same-class and other-class distances are both about 0.99 on MNIST).

5 Scorecard

Table 1: Draft scorecard. "Role": necessary (N), sufficient (S), contributes (C), not supported (–). "Kind": inevitable (I) or avoidable (A).

Hypothesis	Role	Kind	Evidence so far
H1 evidence spread over many inputs	N, S	I	flip size $\propto 1/\sqrt{k}$ exactly; best rule has it; absent under ℓ_2
H2 small, not unreliable	N, S	I	flip size $\propto$ size; reliability irrelevant. Adversarial changes alone teach a fresh network the true classes (10 of 10 runs)
H4a dense signal	S	I	same mechanism as H1
H4a ignoring irrelevant inputs	C	A	network up to 2× worse than best rule when signal is sparse
H4b weight sparsity	fix	A	removes fragility at an accuracy cost
H4c activity sparsity	–	–	no effect; fails all at once
H8 noise sensitivity	restatement of H1	I	gap $\propto \sqrt{k}$
H10 objective	C	A (needs new objective)	ceiling 0.75 vs standard 0.45 at $\epsilon = 0.3$
H7 optimiser bias	C	A	matters when signal sparse and data scarce (real gradient descent checked)
H6 data quantity	–	–	no effect under standard training here
H5 off-manifold	C (small)	A	vanishes with weight decay
H3 shape vs texture	C (via spread)	I (spread), A (architecture)	best rule: flip size × spread constant, frequency irrelevant; convolutional network weak on fine patterns, shape-biased at equal strength
H9 architecture	C	A	right invariance: 4–5× larger flip size than a fully connected net, still 1.6× below the hand-built filter

6 Discussion (draft)

In the toy worlds, the first-order cause of fragility to small per-input changes is inevitable: the evidence a good classifier uses is spread over many inputs or carried by clues that are small compared with the allowed change, and a per-input budget lets the attacker push all of it together. Several explanations that look different (noise sensitivity, dense signals, texture reliance as a category) are the same mechanism seen from other sides: only how widely the evidence is spread matters for the best rule. Avoidable causes exist and can be sizeable (up to a factor of two from ignoring irrelevant inputs; 1.6× from a learned rather than designed filter; a convolutional network that reads fine patterns badly; large when the objective ignores robustness), but they sit on top of the inevitable

one and cannot remove it. This is a hypothesis about large networks, not a result: the real-data checks so far support the geometry but not the size of the effect.

What would change our mind. A real task where the classes are well separated in ℓ_∞, a robust classifier is easy to write down, and standard training still finds a fragile one, would be a case where the avoidable part dominates. Finding or constructing such a case on real data is the most valuable open test.

7 Limits

Independent Gaussian clues, small networks, one-step and projected attacks, no adversarial training, and the real-data confirmation still pending. Weight decay is on by default and affects H5 and H6. "Root cause" is strong language: we claim a map of causes and their rough size in controlled worlds.

A Settings and code map

Toy task: independent Gaussian clues, mean $(y - \frac{1}{2})\Delta\mu_i$, standard deviation σ_i, unclipped; reliable clue $\Delta\mu = 1.0$, $\sigma = 0.3$; weak clues $\Delta\mu = 0.4$, $\sigma = 0.25$. Networks: one hidden layer of 8 units (32 in H5), Adam with learning rate 0.02, 500 full-batch steps, weight decay 10^{-2} unless stated, 4000 training and 2000 test points, 10 runs. Flip size: exact for linear rules, per-point bisection over a 20-step attack for networks. Code: `study/toy_experiments.py` (H4a, H5–H8, H10), `study/shape_texture.py` (H3), `study/invariance.py` (H9), the H2 transfer test (`run_h2b`) and the gradient-descent check for H7 (`run_h7gd`) in `study/toy_experiments.py`, `study/pilot_gap.py`, figures from `study/paper_figures.py`; H1, H2, H4b–c and the real-data checks use the earlier scripts in the repository root.

References

[1] I. Goodfellow, J. Shlens, C. Szegedy. *Explaining and Harnessing Adversarial Examples.* ICLR, 2015.

[2] A. Ilyas, S. Santurkar, D. Tsipras, L. Engstrom, B. Tran, A. Madry. *Adversarial Examples Are Not Bugs, They Are Features.* NeurIPS, 2019.